

Thermal and Radiant Observations

Keep three copied sources distinct while finding hazards

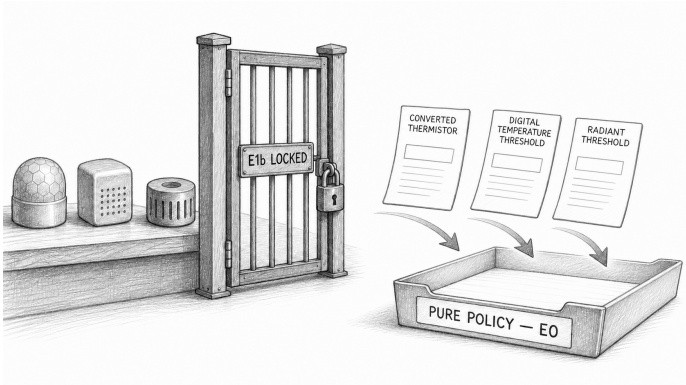
E0 SYNTHETIC REPLAY — DO NOT WIRE OR ENERGIZE A SENSOR

Lesson	062 — component	Time	150–210 minutes
Prerequisites	status, supplied time, copied evidence	Board	compile-only Mega replay
Input	one three-source copied envelope	Observation	named memory result cell

Responsibility. `ThermalRadiantObservationPolicy` preserves an already-converted thermistor result, a distinct Digital Temperature threshold result, and a raw radiant threshold result. It validates and ages each source independently, then reports uncertainty, agreement, saturation, and hazard state. It owns no sensor, pin, transport, conversion curve, clock, or stimulus.

1 Begin after acquisition, not before it

The inventory names Thermistor, Digital Temperature, and Flame families, but those labels are not electrical identities. Lesson 062 does not infer a pinout, protocol, voltage, polarity, unit, or transfer curve from them. Instead, synthetic producers copy three role-specific records across an E0 boundary. The policy evaluates those records without touching hardware.



Alternative text: a pencil drawing places three unidentified physical specimens behind a locked E1b gate, while three copied cards—one converted thermistor, one Digital Temperature threshold, and one radiant threshold—enter a pure policy on the E0 side.

E0 may establish	E0 does not establish
deterministic interpretation of copied records	a module identity, adapter, or wiring diagram
role-specific provenance, sequence, and age	an ADC, GPIO, bus, timer, or supply claim
fixed uncertainty and threshold rules	a physical thermistor conversion curve
bounded radiant candidate state	flame, irradiance, temperature, or life-safety detection

There is no formal electrical schematic in this lesson. An authoritative schematic requires exact specimens, primary evidence, measured rails and currents, proved threshold polarity, and safe-state observations. Those E1b requirements remain open.

2 Copy three noninterchangeable records

The thermistor producer supplies milli-degrees Celsius and an uncertainty. The unidentified Digital Temperature producer supplies only a categorical **Below** or **AtOrAbove** state plus a raw value. The radiant producer supplies the same shape of categorical evidence, but its raw value is unitless and its meaning remains radiant-threshold behavior.

COPIED EVIDENCE ENVELOPE					
THERMISTOR CONVERTED TEMPERATURE		DIGITAL TEMPERATURE (NOT 18B20)		RADIANT THRESHOLD	
SOURCE ID	17	SOURCE ID	84	SOURCE ID	99
CONFIG REV	3	CONFIG REV	7	CONFIG REV	5
CAL REV	5	CAL REV	2	CAL REV	4
SEQUENCE	42	SEQUENCE	87	SEQUENCE	128
OBS TIME	143215	OBS TIME	143215	OBS TIME	143215
SATURATED	NO	SATURATED	NO	SATURATED	NO
PRODUCER STATUS	OK	PRODUCER STATUS	OK	PRODUCER STATUS	OK
VALUE	24736 milli-Celsius	RAW	413	RAW	161 unitless
UNCERTAINTY	85 milli-Celsius	THRESHOLD STATE	Below	THRESHOLD STATE	AtOrAbove

Alternative text: three pencil-drawn cards share fields for source, configuration, calibration, sequence, observation time, saturation, and producer status; only the thermistor card has milli-Celsius and uncertainty, while the other two have raw and threshold state.

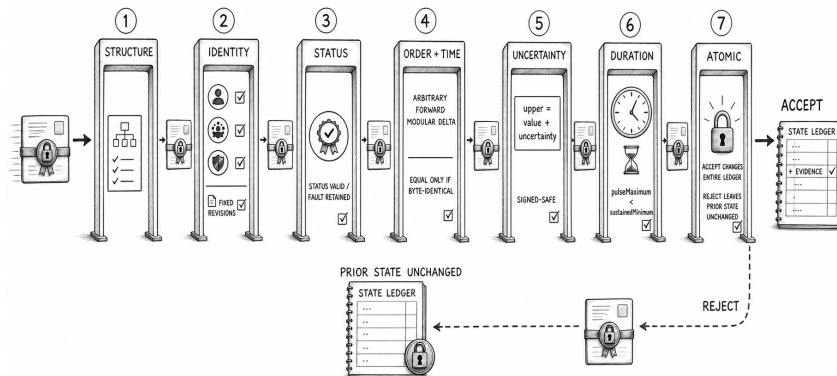
All three source IDs must be nonzero and mutually distinct. Each role retains its own nonzero configuration and calibration revisions, nonzero sequence, observation time, saturation flag, and producer status. The complete envelope is copied; the policy retains no caller pointer. A marketplace family name is not accepted as provenance.

Record	Honest E0 interpretation
converted thermistor	temperature interval supplied by a synthetic producer
Digital Temperature	categorical threshold state; no protocol or temperature unit
radiant	unitless threshold activity; no flame or calibrated radiant unit

Explain why combining the two categorical cards into one generic “hot sensor” would lose evidence: _____.

3 Validate atomically before voting

Configuration requires the warning temperature below the alarm temperature, a wrap-safe maximum age, and nonzero wrap-safe radiant durations with pulse maximum < sustained minimum. Every accepted envelope must preserve fixed role identities and revisions.



Alternative text: a hand-drawn inspection line checks structure, three distinct identities, revisions, producer status, independent sequence and modular time, signed uncertainty arithmetic, and radiant duration configuration before a sealed observation card may change.

Equal sequence and observation time are allowed only when the complete envelope is byte-identical to the accepted envelope. That duplicate is idempotent. The same byte-identical envelope at a valid forward policy time is also idempotent: it neither ages the published snapshot nor extends a radiant candidate. A changed duplicate, regression, backward policy time, exact half-range ambiguity, or identity/revision change rejects the whole update without changing the last observation or radiant candidate. Sequence zero is invalid. Once an accepted role reaches `UINT32_MAX`, that role may remain byte-identical while a companion role advances. Attempting to change the exhausted role returns `CapacityExceeded`; only `reset()` begins a new volatile sequence epoch. Predict the result when the radiant record is malformed but the two thermal records look normal: status _____; observation changes? _____.

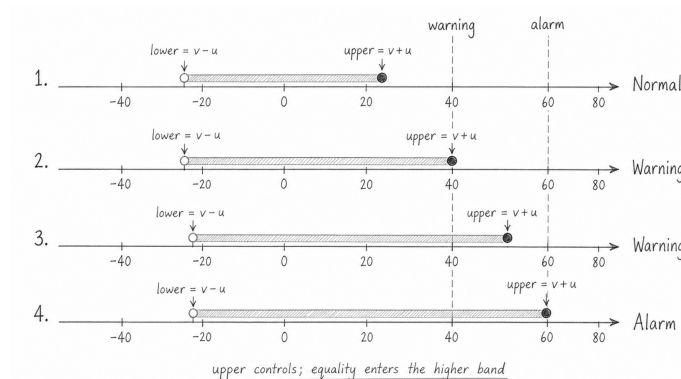
4 Classify a conservative temperature interval

Only the thermistor path carries temperature units. Its supplied value v and uncertainty u define explanatory bounds with widened signed arithmetic:

$$\text{lower} = v - u, \quad \text{upper} = v + u.$$

The policy computes only the upper bound because it alone controls whether the copied interval could reach a hazard. Tests use the lower endpoint as an independent signed-extreme oracle:

upper < warning **Normal**
 upper ≥ alarm **Alarm**
 upper ≥ warning **Warning.**



Alternative text: pencil number lines show temperature uncertainty intervals ending below warning, touching warning, lying between warning and alarm, and touching alarm; equality moves conservatively into the higher quality band.

Value	Uncertainty	Upper bound	Thermistor result
24,000	500	24,500	depends on configured warning
29,000	1,000	30,000	alarm if alarm is 30,000
28,900	1,099	29,999	below that alarm boundary

With warning = 25,000 and alarm = 30,000, classify value 24,000, uncertainty 1,000: _____. Then classify value 29,000, uncertainty 1,000: _____.

5 Preserve agreement without suppressing hazard

The Digital Temperature mapping is fixed and categorical: **AtOrAbove** means alarm and **Below** means normal. It supplies no temperature value. A categorical alarm makes the combined thermal result **Alarm**. A categorical normal paired with a thermistor warning or alarm produces **Disagreement**; if the thermistor interval reaches alarm, **thermalHazard** remains true.

	DIGITAL BELOW	DIGITAL AT OR ABOVE
THERMISTOR NORMAL	NORMAL HAZARD: NO	ALARM HAZARD: YES
THERMISTOR WARNING	DISAGREEMENT WARNING STAYS VISIBLE	ALARM HAZARD: YES
THERMISTOR ALARM	DISAGREEMENT HAZARD: YES	ALARM HAZARD: YES

○ = DISAGREEMENT NEVER VOTES AWAY THE THERMISTOR RESULT
 ✓ = ALARM HAZARD REMAINS RETAINED
 HAZARD RETAINED

Alternative text: a pencil matrix compares thermistor normal, warning, and alarm rows with Digital Temperature below and at-or-above columns; disagreement cells preserve the thermistor hazard flag instead of voting it away.

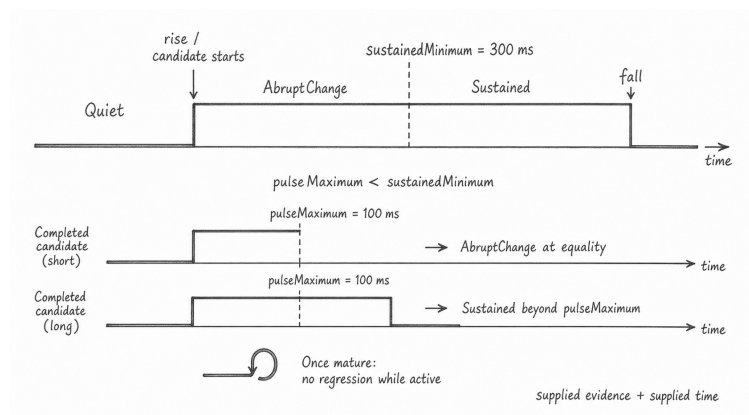
Thermistor	Digital state	Combined quality	Hazard rule
normal	below	Normal	false
normal	at-or-above	Alarm	true
warning	below	Disagreement	warning remains visible
alarm	below	Disagreement	true; disagreement cannot suppress it
warning/alarm	at-or-above	Alarm	true

Stale, saturated, or producer-fault evidence never votes. Its side-specific quality remains visible rather than being averaged into an attractive normal result. Predict the safe interpretation of an alarm thermistor interval paired with stale categorical evidence:

When several copied producers report non-OK status together, the fixed status precedence is thermistor, then Digital Temperature, then radiant. This selects one deterministic reported status; it neither verifies a producer nor lets any copied fault mutate policy state.

6 Track a radiant candidate with supplied time

Radiant threshold activity is temporal evidence, not temperature or flame evidence. An inactive-to-active edge starts a candidate at its accepted observation time. While active, elapsed time below the sustained minimum is **AbruptChange**; equality and later are **Sustained**.



Alternative text: a pencil timing strip shows inactive, active-start, short abrupt activity, equality at the sustained threshold, continued sustained activity, and an inactive edge; a second strip shows a short completed pulse and a longer completed candidate.

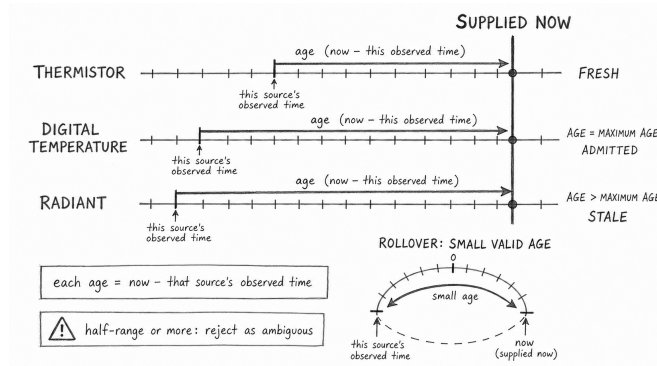
On an active-to-inactive edge, total elapsed time at or below the pulse maximum reports **AbruptChange**; a longer completed candidate reports **Sustained**. A byte-identical duplicate cannot extend duration. Ambient saturation reports **SaturatedAmbient** and clears the candidate, so later evidence must begin a new edge.

Boundary	Expected radiant quality
active for sustained minimum minus one tick	AbruptChange
active for exactly sustained minimum	Sustained
inactive edge at exactly pulse maximum	AbruptChange
inactive edge one tick beyond pulse maximum	Sustained
ambient saturation	SaturatedAmbient ; candidate cleared

If pulse maximum is 100 ms and sustained minimum is 300 ms, predict the result of an inactive edge after 101 ms: _____. Explain why a duplicate at 250 ms cannot turn a 200 ms candidate into 250 ms of evidence: _____.

7 Age each role independently

TimePoint now is the only clock. The policy computes thermistor, Digital Temperature, and radiant age separately with modular unsigned time. Evidence exactly at maximum age is admitted; one tick older is stale. This prevents a fresh source from laundering an old companion into a simultaneous measurement.



Alternative text: three pencil clock tracks end at one supplied now marker but begin at different observation times; one is fresh, one sits on the inclusive maximum-age boundary, and one is stale, with a rollover arc shown separately.

Complete the diagnosis:

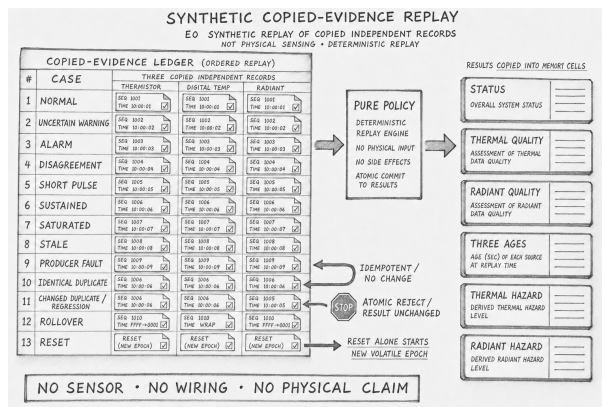
1. thermistor age equals maximum age: _____;
2. Digital Temperature age is one tick greater: _____;
3. radiant time difference is exactly the modular half range: _____;
4. thermistor crosses ordinary unsigned rollover with a small valid age: _____;
5. reset then snapshot: quality _____, candidate active? _____.

8 Replay without stimulus or wiring

The Mega example constructs synthetic envelopes, supplies time, updates the policy, and copies snapshots into a named volatile result cell. It acquires no ADC channel, pin, timer, bus, endpoint, display, or supply. Serial is not the proof, and the compile-only replay makes no physical observation claim.

```
ThermalRadiantObservationPolicy environment(config);
environment.initialize();

envelope.thermistor.sequence += 1U;
envelope.digitalTemperature.sequence += 1U;
envelope.radiant.sequence += 1U;
const Status updateStatus = environment.update(now, envelope);
const ThermalRadiantObservation result = environment.snapshot();
```



Alternative text: a pencil replay ledger feeds normal, uncertain-warning, alarm, disagreement, short-pulse, sustained, saturated, stale, producer-fault, duplicate, regression, rollover, and reset envelopes into one policy and records status, both qualities, three ages, and two hazard flags in named result cells.

The deterministic proof must cover signed extrema; uncertainty immediately below, at, and across both thermal thresholds; every categorical agreement combination; radiant pulse and sustained neighbors; ambient saturation; source collision; independent stale, future, and regressing records; changed duplicates; producer faults; sequence exhaustion; rollover; reset; rejected update atomicity; and byte-stable replay.

9 Separate software evidence from open physical gates

The canonical Mega replay measures 10,076 bytes of flash and 1,133 bytes of static SRAM. The static-SRAM result is an independently accepted target miss against 1,024 bytes and remains below the 1,536-byte hard limit. The exact isolated no-LTO gate measures 5,156 bytes of flash, 279 bytes of static SRAM, 264 bytes of conservative synchronous stack, and a 112-byte policy object. The caller retains 59 bytes of stack and a 76-byte observation, for 135 caller-owned bytes; 7,521 bytes of museum-monitor SRAM remain. These compiler measurements are not runtime, electrical, optical, thermal, or physical evidence.

Gate	E0 pass evidence	Still open
resource acquisition safe state	zero hardware claims and compile-only replay reset and rejection clear or preserve bounded policy state	exact endpoints, rails, current, and ownership de-energized physical outputs and observation path
thermal radiant replay	widened interval and agreement boundary tests pulse, sustained, saturation, and rollover traces byte-stable named result cells	thermistor curve and Digital Temperature identity response, polarity, geometry, and harmless stimulus bench repeatability and physical acceptance

E1b remains open. Qualify the exact thermistor, the distinct Digital Temperature specimen, and the radiant module independently before publishing an adapter, wiring, physical units, or threshold polarity. Any radiant stimulus must be an owned, documented harmless low-energy IR target/source pair or an owned TV remote used only as a stimulus: attended, never aimed at eyes, active for at most 5 seconds, followed by at least 30 seconds inactive, and never replayed. Lasers, flame, heaters, hot wire, combustible targets, ignition, continuous or unattended exposure, and unknown targets are excluded.

E1b also requires a non-Serial observation path plus separate proof of resource acquisition and physical safe state. Lesson 062 contains no powered acceptance record and makes no life-safety, fire-detection, preservation, or absence-of-hazard claim.

10 Completion record

Check	Result or evidence
Three distinct roles, identities, revisions, and copied fields	
Signed widened uncertainty and both threshold neighbors	
Every thermal/categorical agreement and hazard result	
Radiant pulse, sustained, saturation, and candidate reset	
Independent age, duplicate, regression, exhaustion, and rollover	
Atomic rejection, reset, replay, and named result cells	
Zero-hardware E0 boundary and all open E1b work	

Completion statement: I can distinguish converted temperature from categorical threshold and radiant evidence, apply conservative uncertainty and disagreement rules, reproduce every radiant timing boundary, and name the electrical and physical claims that remain open.

Learner: _____ Reviewer: _____ Date: _____

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